Question ID: 5a74f696

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question
An investment account was opened with an initial value of $\$890$. The value of the account doubled every 10 years. Which equation represents the value of the account $M(t)$, in dollars, t years after the account was opened?

Answer
A. $M(t)=890\left(\frac{1}{2}\right)^{\frac{t}{10}}$

B. $M(t)=890\left(\frac{1}{10}\right)^{\frac{t}{2}}$

C. $M(t)=890(2)^{\frac{t}{10}}$

D. $M(t)=890(10)^{\frac{t}{2}}$

Correct Answer: C

Rationale
Choice C is correct. It's given that t represents the number of years since the account was opened. Therefore, $\frac{t}{10}$ represents the number of 10-year periods since the account was opened. Since the value of the account doubles during each of these 10-year periods, the value of the account can be found by multiplying the initial value by $\frac{t}{10}$ factors of 2. This is equivalent to $2^{\frac{t}{10}}$. It's given that the initial value of the account is $\$890$. Therefore, the value of the account $M(t)$, in dollars, t years after the account was opened can be represented by $M(t)=890(2)^{\frac{t}{10}}$.

Choice A is incorrect. This equation represents the value of an account if the value of the account halves, not doubles, every 10 years.

Choice B is incorrect. This equation represents the value of an account if the value of the account decreases by 90%, not doubles, every 2, not 10, years.

Choice D is incorrect. This equation represents the value of an account if the value of the account increases by a factor of 10, not doubles, every 2, not 10, years.